

What is this project about?

We are building a computer vision program that can look at a fingerprint image and guess if it belongs to a male or a female. This is highly useful for police forensics to narrow down suspects, and for advanced security systems.

The Dataset

We are using the SOCOFing (Sokoto Coventry Fingerprint) dataset. It contains 6,000 fingerprint images of different people. It is a great dataset because the images are real, clear, and perfect for training computer vision models.

How we cleaned and prepared the data

Raw images cannot be fed directly into a machine learning model. We built a simple Python pipeline to prepare them:

- Finding the Labels: The gender (Male/Female) was not in a separate document; it was written inside the name of each image file (like 1__M_Left_index.BMP). We wrote a script to read the filenames, find the "M" or "F", and convert them into simple numbers (0 for Male, 1 for Female).
- Color & Size Fix: Fingerprints do not need color to be identified. We converted all images to black-and-white (grayscale) and resized them to a standard 96x96 pixels so they are all exactly the same size.
- Scaling the Numbers: Computer screens show pixels using numbers from 0 to 255. We divided all these numbers by 255.0 to scale them down between 0.0 and 1.0. This makes it much easier and faster for our AI model to learn.
- Splitting the Data: We split our dataset into two parts: 80% for training our model, and 20% for testing it later. This ensures we do not test the model on images it has already seen.
- Saving the Work: We packed the finished, clean data into a single compressed file called `socofing_gender_preprocessed.npz` so we can load it instantly in our next milestone.

Tools Used

We used Python along with three main packages: OpenCV (to load and resize images), NumPy (to organize the images into clean grids of numbers), and Scikit-Learn (to split our data into training and testing sets).